

MAHNOOR KHALID

Full Stack Developer & AI Engineer

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PROFESSIONAL SUMMARY

Highly motivated Full Stack Developer and AI Engineer with expertise in building robust web applications and integrating advanced Artificial Intelligence systems. Proficient in modern JavaScript/TypeScript frameworks like Next.js, backend architecture with Python/FastAPI, and developing autonomous Agentic AI workflows. Proven ability to transition complex logic into production-ready software solutions. Committed to continuous learning, collaborative innovation, and leveraging cutting-edge LLMs to solve real-world problems.

TECHNICAL EXPERTISE

- **Languages & Core Web:** TypeScript, JavaScript, Python, HTML5, CSS3, SQL
- **Frontend Frameworks & Styling:** React.js, Next.js (App Router), Tailwind CSS, Responsive UI/UX Design, Docusaurus
- **Backend & API Development:** Node.js, FastAPI, JWT Authentication
- **AI & Large Language Models:** Generative AI, Agentic AI, Chainlit, Streamlit, Prompt Engineering, Context Engineering, API Integration (Claude, Gemini, Qwen, speckit plus)
- **Databases & Headless CMS:** PostgreSQL, Sanity CMS
- **Developer Tools & Platforms:** Git, GitHub, Docker, Vercel, npm/pnpm, Postman

TECHNICAL PROJECTS & EXPERIENCE

AI & Full Stack Developer (Independent Projects) | 2024 – Present

Personal AI Multi-Platform Employee Agent

- Designed and deployed an autonomous personal AI assistant capable of managing automated workflows across major platforms including Gmail, WhatsApp, LinkedIn, Facebook, Instagram, and Twitter.
- Utilized **Python**, **Next.js**, **Typescript**, **Kubernetes**, and **Docker** to implement intelligent conversational memory and agentic decision-making logic.
- Integrated various platform-specific REST APIs to allow the agent to read, synthesize, and draft context-aware responses seamlessly.

Physical AI Humanoid Robotics Educational Portal

- Engineered a high-performance documentation and learning portal using the **Docusaurus** framework, incorporating interactive AI study assistants.
- Developed a secure backend utilizing **Node.js** and **JWT (JSON Web Tokens)** for seamless user registration, session management, and authentication.
- Embedded a fine-tuned chat assistant powered by **Gemini APIs** to guide students through complex robotics and physics concepts in real-time.

AI-Powered Task Optimization Application

- Built a modern full-stack Todo application using **Next.js**, **Typescript**, and **Tailwind CSS** for a responsive, clean user experience.

- Integrated a local AI helper utilizing **FastAPI** to automatically prioritize, categorize, and schedule user tasks based on natural language input.
- Secured application routes using JWT authentication and stored structured task data in a relational **PostgreSQL** database.

Headless E-Commerce Watch Platform

- Developed a responsive, modern e-commerce storefront for watches using **Next.js** and styled with **Tailwind CSS** for fluid responsiveness.
- Implemented **Sanity CMS** as a headless backend, enabling content creators to dynamically manage inventories, specifications, and pricing without redeploying code.
- Optimized search engine visibility (SEO) and static site generation (SSG) speeds, deploying the final build to **Vercel** for global edge delivery.

EDUCATION

- **Governor House Initiative (GIAP)**
Specialization in Generative AI, Agentic AI, & Cloud-Applied Computing | Ongoing
- **University of Karachi**
Masters of Science in Biochemistry | 2023 (First Division)
- **University of Karachi**
Bachelors of Science in Biochemistry | 2022 (First Division)
- **Govt. Degree College (Sheikh Zaid Center)**
Intermediate (Pre-Medical) | 2019 (Grade A)
- **Institute for Learning & Development (ILAD)**
Matriculation (Science Group) | 2017 (Grade A1)

ADDITIONAL WORK EXPERIENCE

Lead / Assistant Educator | Various Academic Institutions | 2019 – 2024

Smart School | Happy Palace Group of Schools | Little Folk's School

SOFT SKILLS & COMPETENCIES

- **Communication:** Technical Documentation, Project Reporting, Team Syncs
- **Cognitive:** Creative Problem Solving, Rapid Upskilling, Analytical Thinking
- **Collaboration:** Version Control (Git/GitHub), Time Management